Chapter 8 · Solutions

P37 · Identify the scale before using arithmetic

Hint. A numeric storage code is not automatically a quantitative measurement.

Department code is nominal; Low/Medium/High is ordinal; Celsius is interval; mass in kg is ratio. Mean department codes do not have a department interpretation. Ordinal labels have order without guaranteed equal gaps.

For x=60, min-max to [0,1] is (60−20)/(80−20)=2/3. To [-1,1], use −1+2×(2/3)=1/3. Z-score is (60−50)/10=1.

Decimal scaling needs 80/10^j<1. j=1 fails; j=2 works. The value becomes 60/100=0.60.

If all observations are 7, max−min=0 and SD=0. Ordinary min-max and z-score formulas divide by zero. A handling rule must be supplied; the formulas do not decide it. Relational normalization is unrelated to this numerical scaling.

P38 · Follow the boundary convention

Hint. Sort and assign bins before calculating replacement values.

Method	Bin memberships	Bin means	Smoothed sequence
equal width	{0,1,2}; {3}; {8,9}	1; 3; 8.5	1,1,1,3,8.5,8.5
equal frequency	{0,1}; {2,3}; {8,9}	0.5; 2.5; 8.5	0.5,0.5,2.5,2.5,8.5,8.5

3 belongs in [3,6), not [0,3). 9 belongs in the final closed interval. Equal-frequency bins use two observations each, not equal numerical widths.

Bin labels produce categorical membership labels in place of these numerical means. Neither labels nor mean smoothing generally permit exact recovery of every original value.

P39 · Population weights differ from sample allocation

Hint. Weight stratum means by their population shares.

The population-mean estimate is (90/100)×10+(10/100)×30=12. The unweighted observed sample mean is (5×10+5×30)/10=20. Equal sampling from unequal strata does not give equal population weights.

Two ordered draws without replacement have 5 choices followed by 4: 20 outcomes. With replacement, both draws have 5 choices: 25 outcomes. Unordered subsets would be a different counting problem.

The first-stage unit in the stated cluster design is a village. Including all its households does not turn the design into independent random household draws.

P40 · Count overhead and demand reconstruction

Hint. Count the actual encoded representation, not only its payload.

Encoded size=600+100=700 bytes. The original/encoded ratio is 1000/700=10/7≈1.4286. Saving=(1000−700)/1000×100%=30%.

AAAABBCD has runs (A,4),(B,2),(C,1),(D,1). Four 16-bit pairs plus an 8-bit header cost 72 bits. The original eight 8-bit symbols cost 64 bits. The encoding expands by 8 bits, or 12.5%. Run-length encoding is not automatically shorter.

There are four two-bit inputs: 00,01,10,11. Only three strings are shorter than two bits: the empty string,0,1. Exact reconstruction requires different inputs to have distinguishable encodings. Four inputs cannot all fit into three distinct shorter encodings. Thus a lossless code cannot shorten every two-bit input.